Residential Broadband Analytics

CUSTOMER CHURN PREDICTION

Determine Machine Learning model with high ROC-AUC

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INTRODUCTION

BUSINESS BACKGROUND

Telkom Indonesia's **Residential Broadband business** serves residential customer across **DKI Jakarta**. As competition intensifies, customer retention becomes an important business priority, as **retaining existing customers is more cost-effective than acquiring new ones**. Therefore, Telkom aims to proactively identify customers at risk of churn and take timely retention actions before they leave

BUSINESS PROBLEM

How can Telkom identify the **20% of customers most likely to churn who potentially account for 80% of recurring revenue**, enabling the business to prioritize retention efforts where they matter most?

BUSINESS OBJECTIVE

Develop a **Machine Learning model** that predicts customer who are likely to churn, identifies the key actionable factors associated insights to support proactive customer retention.

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INTRODUCTION

BUSINESS QUESTION

- Who should the business prioritize?
- Why are these customer groups at risk?
- Which group should receive retention campaigns?
- How can the model support proactive retention?

BUSINESS IMPACT

If churn isn't addressed, the business can potentially face:

- Customer/Revenue loss
- Lost future revenue opportunity
- Higher effort required to acquire replacement customers
- Missed opportunities to intervenes before cancellation.

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EXECUTIVE SUMMARY

Dataset

Train: 20,000 rows, 29 col
Test: 4,000 rows, 29 col
Target col: Churn

XGBoost is best model to predict churn

CV ROC - AUC:
LogReg : 0.975
RandomForest : 0.976
XGBoost : 0.980
XGBoost (IC) : 0.956

Finding Model Targeted

ChurnProb : 0.1922
Cus. capt : 24.43%
Loss mon : 76.28%
Worth : 161,053,259.19
Acc Risk : 50,091,617

Top 5 Features

Lo avg active days 3m
Lo avg quality 3m
Hi avg resolution days 6m
Lo usage ratio
Hi avg downtime

Targeted Customer Retention

339 Customer

Recommendation

Prioritize active high-risk customers for proactive retention using the model

TABLE OF CONTENT

Data Loading & Data Preparation

- Import Libraries
- Import dataset
- Data Cleaning
- Datetime fixing

Feature Engineering on Train & Test Data

- Extract Several Information
- One hot Encoder Categorical col
- Drops Id & Datetime

Exploratory Data Analysis

- Check Target Distribution
- Check data relationship

Model Development Life Cycle 1

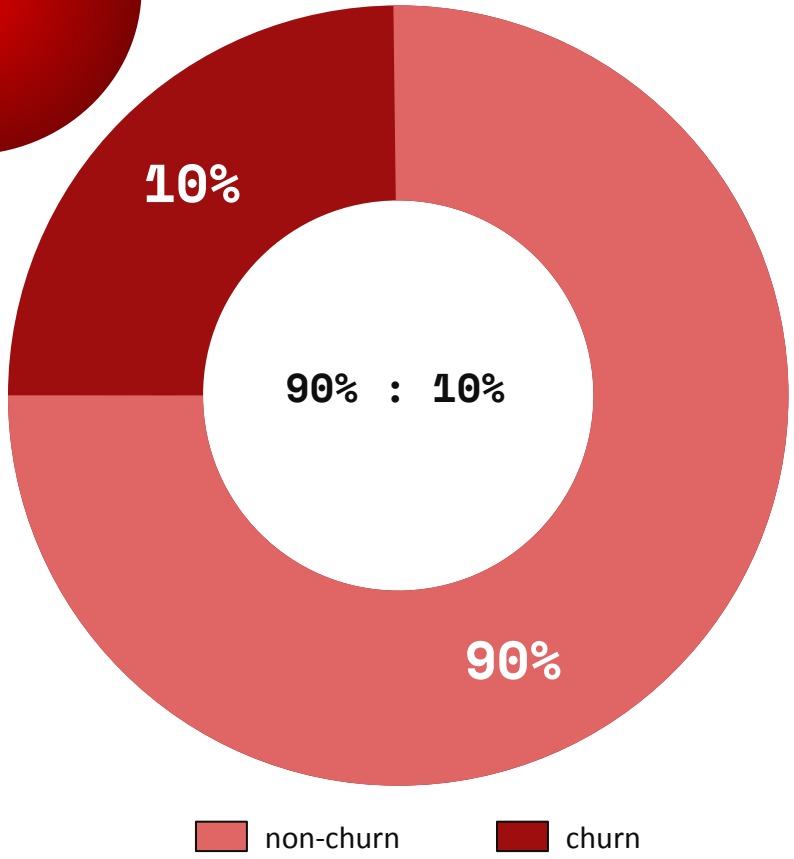
- Model Development
- Model Evaluation
- Model Interpretability (SHAP)

Model Development Life Cycle 2

- XGBoost + Teacher-Student Distillation
- Model Evaluation
- SHAP

Business Recommendations

- Business Recommendation
- Conclusion



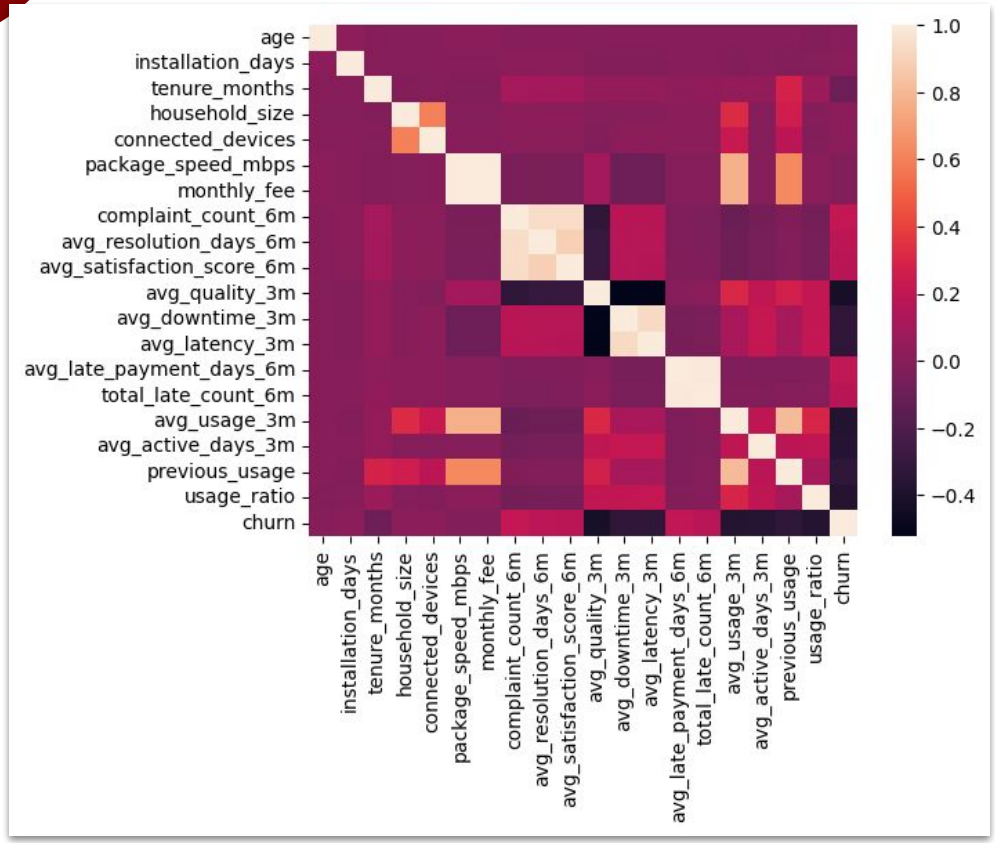
EXPLORATORY DATA ANALYSIS

Churn Represent only **10%**. Creating a **90:10** class imbalance.

This imbalance can cause a model to favor the majority class, churn-specific evaluation and modeling strategies important

Click to go to the notebook

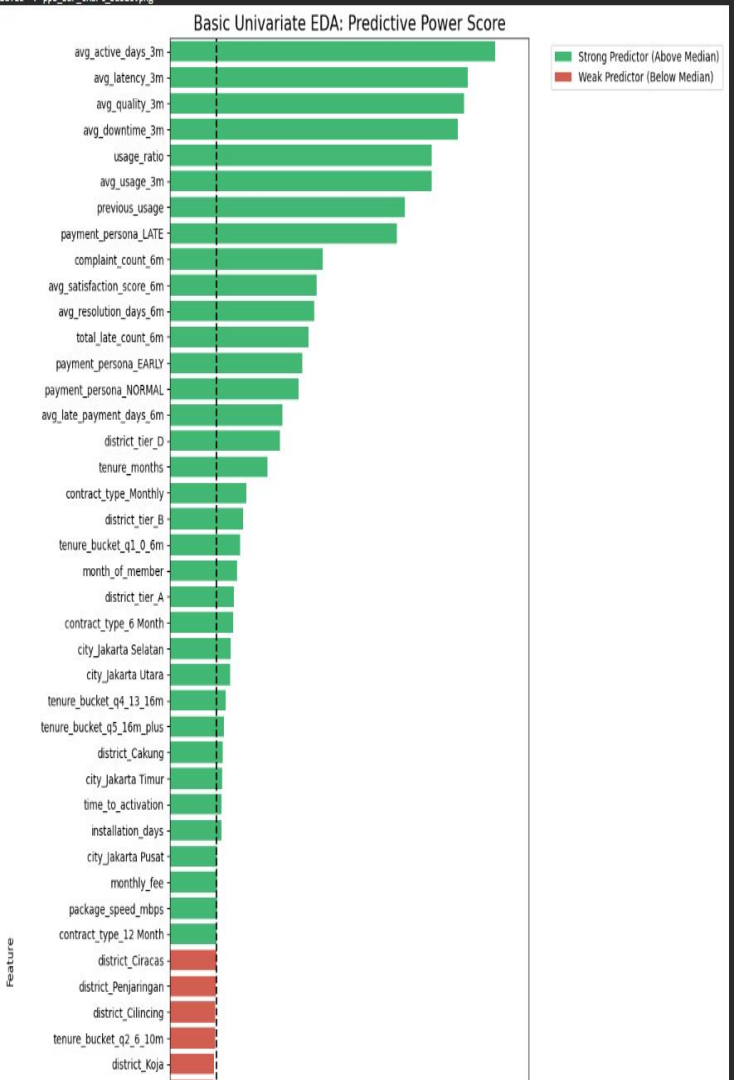
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EXPLORATORY DATA ANALYSIS

Spearman correlation was used to identify monotonic relationship between numerical variables based on their ranks. The Results provide an initial view of variable associations.

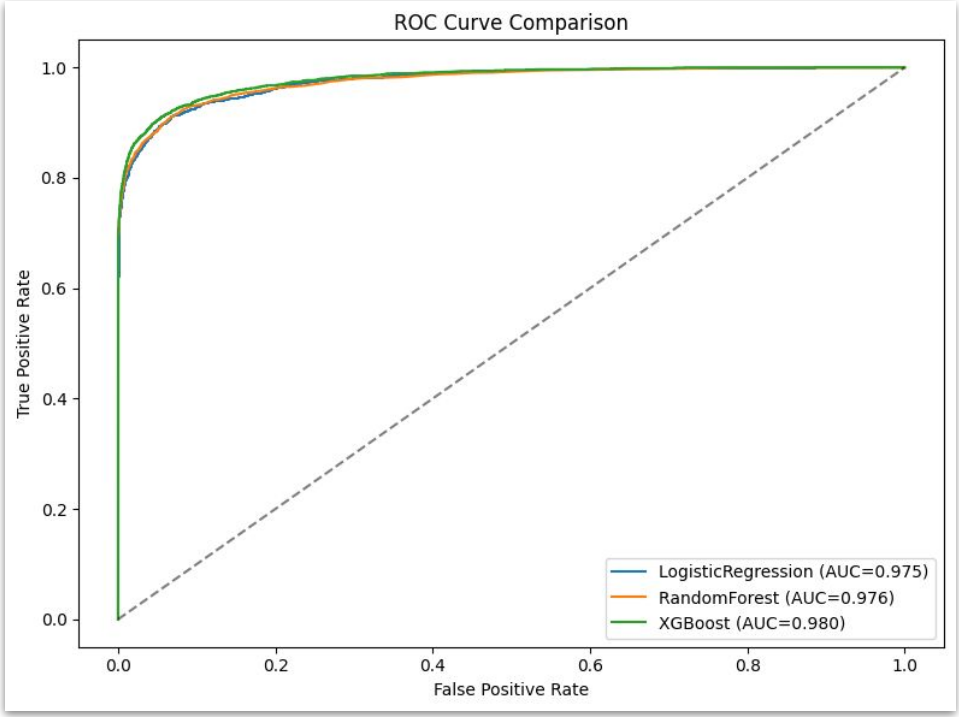
While PPS was used in the next step to capture potentially non-linear relationship that spearman may miss.



PREDICTIVE POWER SCORE (PPS)

PPS measures how well each feature can predict churn individually. Unlike correlation, it can capture non-linear relationship and can be applied beyond purely numerical relationships.

The chart shows that features above the median PPS have relatively stronger predictive power, helping identify variables that may be more relevant for churn modeling.



MODEL DEVELOPMENT LIFE CYCLE MDLC 1

Churn Prediction Models AUC Comparison:

XGBoost - Advance ML Model is win the **AUC = 0.980**

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MODEL DEVELOPMENT, COMPARISON & HYPERPARAMETER TUNING

XGBoost CV ROC-AUC **0.9800**	Precision	Recall	F1 score
0	**0.988**	**0.957**	**0.972**
1	**0.699**	**0.895**	**0.785**

Random Forest CV ROC-AUC **0.9759**	Precision	Recall	F1 score
0	0.979	0.990	0.984
1	0.897	0.808	0.850

Logistic Regression CV ROC-AUC **0.9755**	Precision	Recall	F1 score
0	0.988	0.943	0.965
1	0.634	0.897	0.743

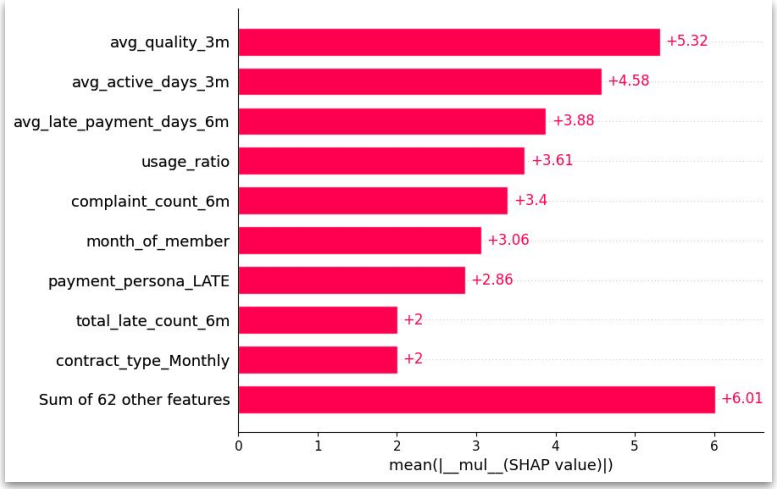
Metrics:

- **ROC-AUC**
  how well the model distinguishes churners from non-churners across different thresholds.
- **Precision**
  Of all customers predicted to churn, how many actually churn?
- **Recall**
  Of actual churners, how many does the model successfully identify?
- **F1 Score**
  A balance between precision and recall.

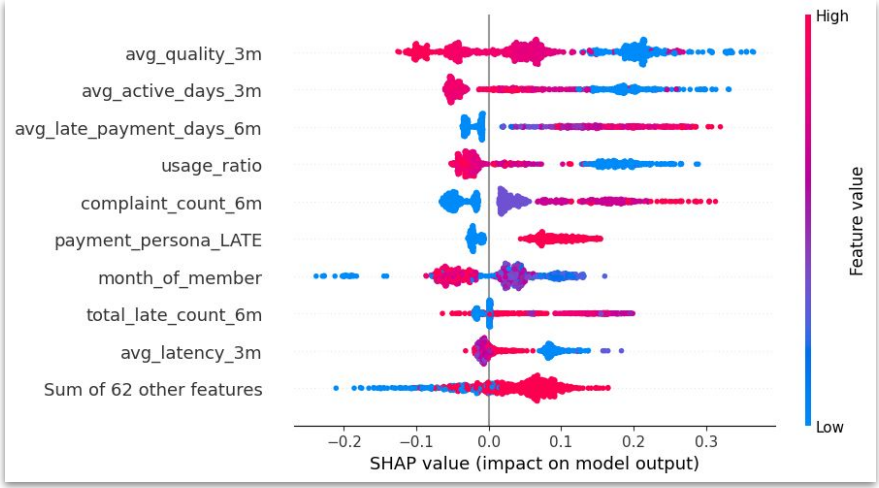
MDLC 1 selected XGBoost based on its highest CV ROC-AUC among the evaluated models

MODEL INTERPRETABILITY (SHAP) V1

Global SHAP bar



SHAP beeswarm



SHAP analysis shows that customer engagement, network quality, payment behavior, and complaint frequency are the key contributions to churn predictions. Lower network quality, lower active days, and lower usage ratio tend to increase the predicted probability days and complaint frequency push the prediction toward churn.

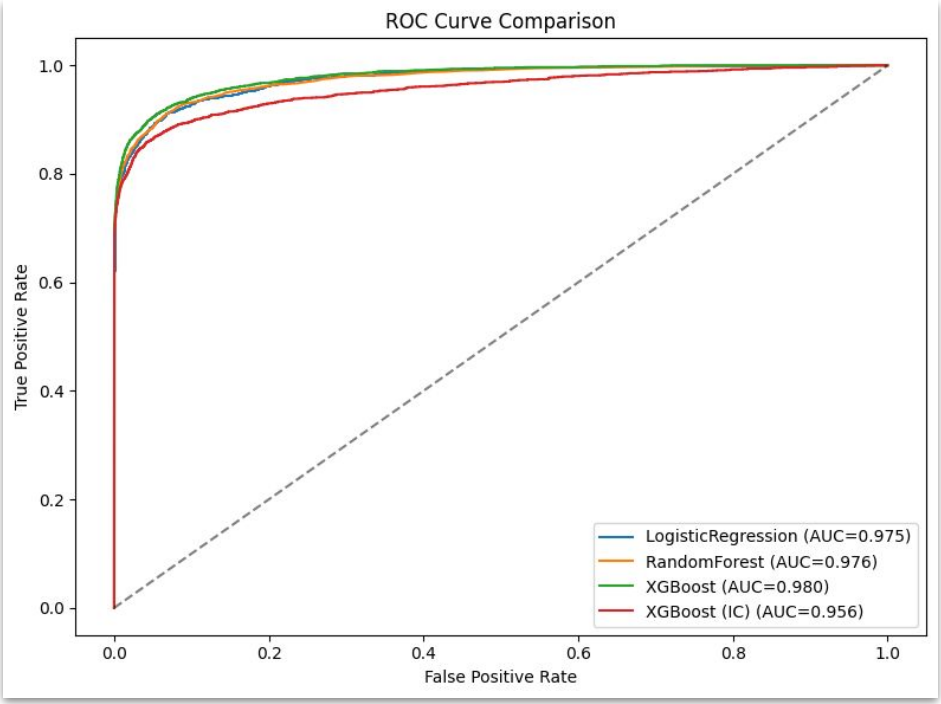
REVENUE PRIORITIZATION

Telkom Does not need to target every customer at risk of churn. By prioritizing 22.15% of customers with a churn probability of >= 21.3%. The model captures 79.48% of the expected monthly recurring revenue loss in the test dataset.

Target Customer Segment	Key Predictive Signals
22.15% of customers Churn probability >= 21.3% Expected MMR Received: IDR 145M (79.48%) Expected MRR at Risk IDR 39M	• **Avg. Late Payment days (6m)** • Avg. quality (3m) • **Complaint count (6m)** • **Total late count (6m)** • Usage ratio • **Month of member** • Avg active days (3m) • LATE-Payment persona • Avg. latency

However, Several key predictive features are **symptoms** rather than directly actionable levers. This raises to develop prediction model (MDLC 2) grouping categorical features into actionable, non-actionable, and symptoms

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MODEL DEVELOPMENT LIFE CYCLE MDLC 2

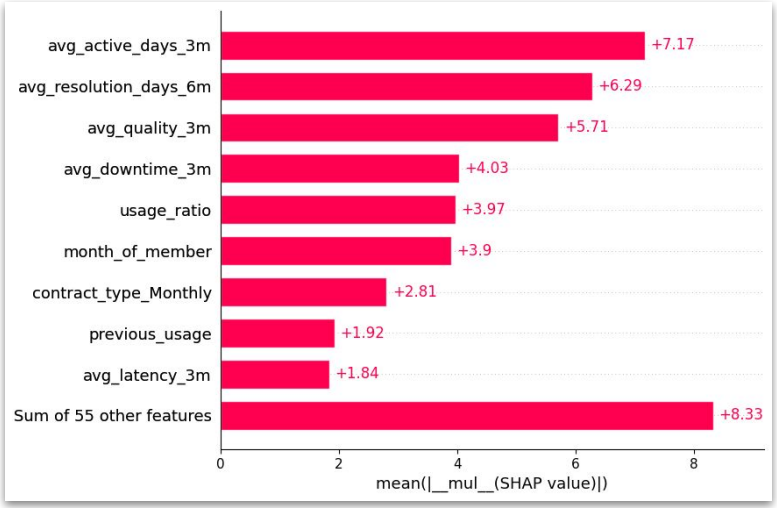
We know already XGBoost is the best model so far, we'll try to improve using in-context learning (IC):

- Grouping feature into ("category", "Actionability")
- XGBoost + Teacher-Student Distillation

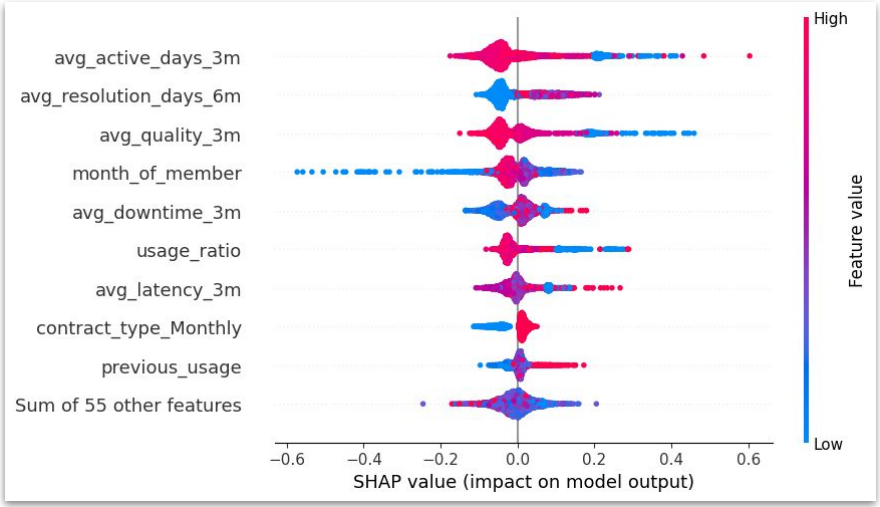
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MODEL INTERPRETABILITY (SHAP) V2

Churn probability	Customer to be cons.	MRR Received 76.28%	MRR Risk
>=19.22%	29.43%	IDR 161M	IDR 50M

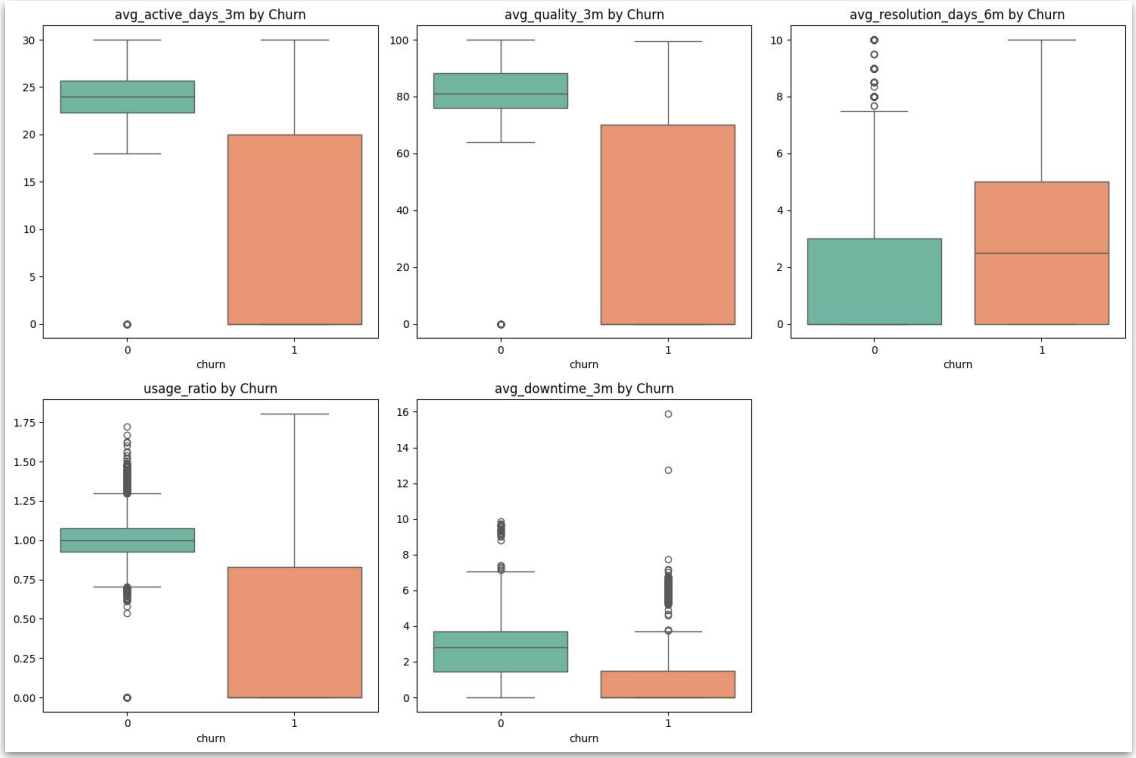


Global SHAP bar



SHAP beeswarm

TOP 5 FEATURE VALIDATING



Confirms top 5 key actionables Key features show distinct distributions between churned and non-churned customers, validating their relevance for churn modeling

HIGH RISK CUSTOMER

Who is our targeted customer?

High Risk customer from test data

customer_id	loss	churn_prob	group
CUST020001	71769.437939	0.220829	month_of_member
CUST020002	249942.019582	0.999768	avg_quality_3m
CUST020003	58951.427042	0.256311	avg_resolution_days_6m
...	...	...	...
CUST023997	229962.409735	0.999837	avg_quality_3m
CUST023998	70094.370246	0.304758	avg_active_days_3m
CUST023999	229963.533878	0.999841	avg_quality_3m

1177 rows

Targeted customer High Risk customer from test data

customer_id	churn_prob	group	monthly_fee
CUST020001	0.220829	month_of_member	325000
CUST020008	0.227068	avg_resolution_days_6m	325000
CUST020016	0.356055	avg_resolution_days_6m	290000
...	...	...	...
CUST023964	0.437913	avg_resolution_days_6m	290000
CUST023967	0.311187	avg_active_days_3m	290000
CUST023981	0.234274	contract_type_Monthly	290000

339 rows/339 customer customer targeted receive retention campaigns

BUSINESS RECOMMENDATION



Network Quality Investment

Prioritize Internet quality, latency and downtime improvements towards customers at network quality group with monthly fee is median.



Revenue-Based Prioritize

Use expected monthly loos-not just risk scores-to prioritize retention outreach, starting with high-value, high-risk customers. Low effort- High Impact.



Resolution Time Reduction

Prioritize recurring or high-risk complaints for faster escalation to reduce resolution time and prevent prolonged service dissatisfaction.



Customer Engagement Recovery

Prioritize low-activity customers with personalized engagement offers, service check-ups, and usage-based outreach to increase active days and reduce disengagement



Long-Term Contract Incentives

Encourage customers to migrate from monthly contracts to 6 month & 12 month plans through targeted incentives, as longer-term contracts show lower churn rates



Early-Warning System

Build an automated monitoring system to flag customers with low usage ratios and sharply declining active days over the past three months.

CONCLUSION

1. Who should the business prioritize?
   - Those with a churn probability >=19.22% and monthly fee >median (considered as HIGH risk)
   - XGBoost model as point of reference we can proactively address 76.28% of our total expected MRR loss (on test dataset)

2. Why are these customer groups at risk?
   Top 5 key drivers:
   - Lo avg, active days 3m
   - Lo avg. quality 3m
   - Hi avg. resolution days 6m
   - Lo usage ratio
   - Hi avg. downtime 3m

3. Which group should receive retention campaigns?
   There are 339 customers who has High risk for churn and total spending above median because investing retention budget in LOW-value churners may have negative ROI. it is stored at variable “targeted_customers_grouped”, complete with their top key drivers.

4. How can the model support proactive retention?
   Score customers weekly/monthly, flagged customers according according to their top feature which induce churn, route them to their respective retention team or automated campaigns orchestration engine.

Thank you!

Keep **Update** with my latest **insights**



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APPENDIX

Dataset Train	[CLICK THE LINK]
Dataset Test	[CLICK THE LINK]
Google Colab - ML Notebook 1	[CLICK THE LINK]
Google Colab - ML Notebook 2	[CLICK THE LINK]
Pitching Recording	[CLICK THE LINK]